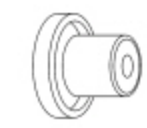


# Valve Cover — RH

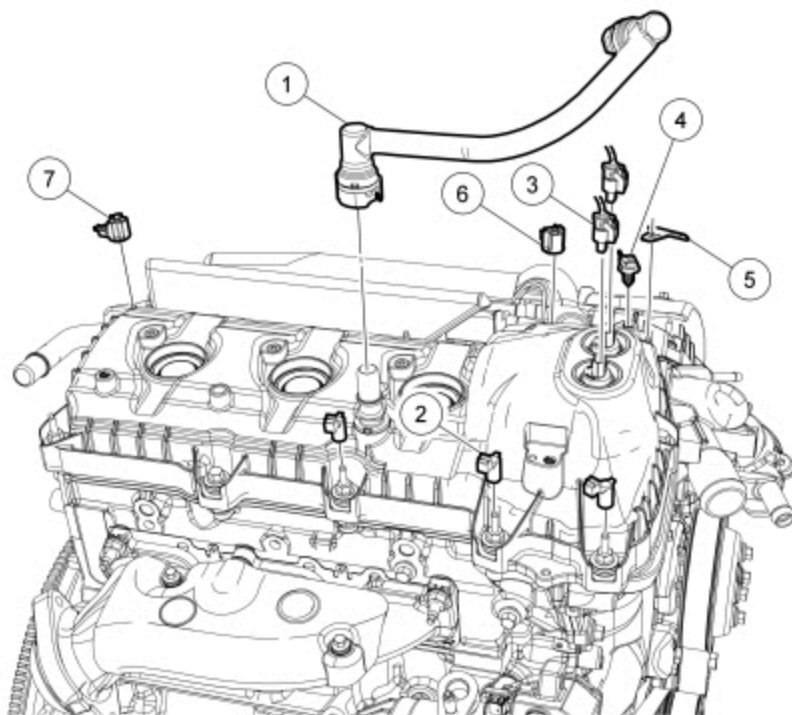
 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

|                                                                                                |                                                               |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <br>ST1326-A   | Handle<br>205-153 (T80T-4000-W)                               |
| <br>ST2212-A   | Installer, Differential Bearing Cone<br>205-142 (T80T-4000-J) |
| <br>ST2983-A  | Installer, Spark Plug Tube Seal<br>303-1247/2                 |
| <br>ST2982-A | Remover, Spark Plug Tube Seal<br>303-1247/1                   |

Material

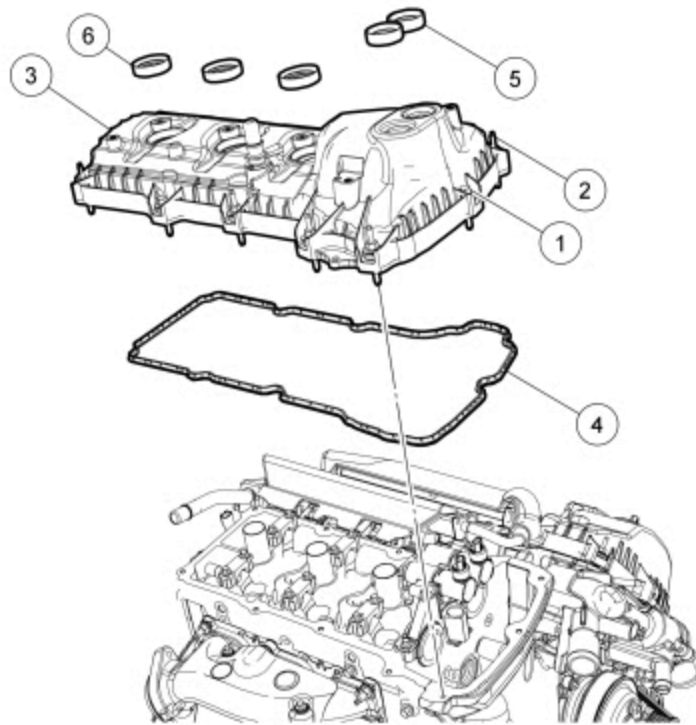
| Item                                                       | Specification |
|------------------------------------------------------------|---------------|
| Motorcraft® High Performance Engine RTV Silicone<br>TA-357 | WSE-M4G323-A6 |
| Motorcraft® Metal Surface Prep<br>ZC-31-B                  | —             |
| Motorcraft® Silicone Gasket Remover<br>ZC-30-A             | —             |



N0114107

| Item | Part Number | Description                                                                          |
|------|-------------|--------------------------------------------------------------------------------------|
|      | —           |                                                                                      |
| 1    | 6A6646A664  | PCV tube                                                                             |
| 2    | —           | RH valve cover stud bolt wiring harness retainer (part of 14A411) (3 required)       |
| 3    | —           | RH Variable Camshaft Timing (VCT) electrical connector (part of 14A411) (2 required) |
| 4    | —           | RH valve cover wiring harness pin-type retainer (part of 14A411)                     |
| 5    | —           | RH valve cover stud bolt wiring harness retainer (part of 14A411)                    |
| 6    | —           | RH valve cover stud bolt wiring harness retainer (part of 14A411)                    |
| 7    | —           | RH valve cover stud bolt wiring harness retainer (part of 14A411)                    |

Valve Cover — RH (View 2 of 2)



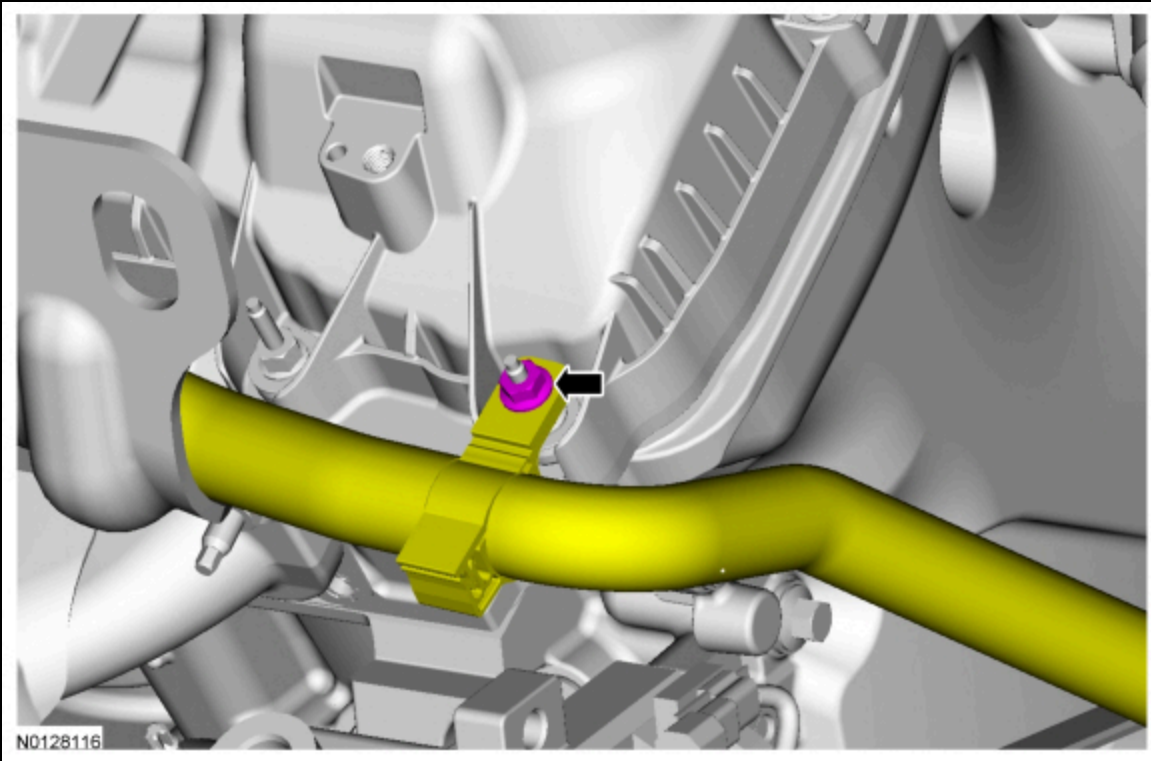
N0114108

| Item | Part Number                 | Description                                               |
|------|-----------------------------|-----------------------------------------------------------|
| 1    | 6C519 <a href="#">6C519</a> | Valve cover bolt (3 required)                             |
| 2    | 6C519 <a href="#">6C519</a> | Valve cover stud bolt (8 required)                        |
| 3    | 6582 <a href="#">6582</a>   | RH valve cover                                            |
| 4    | 6584 <a href="#">6584</a>   | RH valve cover gasket                                     |
| 5    | 6C535 <a href="#">6C535</a> | Variable Camshaft Timing (VCT) solenoid seal (2 required) |
| 6    | 6C535 <a href="#">6C535</a> | Spark plug tube seal (3 required)                         |

## Removal

**NOTICE:** During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces that enters the oil passages, coolant passages or the oil pan, may cause engine failure.

1. Remove the RH ignition coils. For additional information, refer to Section 303-07B.
2. Disconnect and remove the PCV tube quick connect coupling from the RH valve cover. For additional information, refer to Section 310-00.
3. Disconnect the 2 RH Variable Camshaft Timing (VCT) solenoid electrical connectors.
4. Detach all of the wiring harness retainers from the RH valve cover and stud bolts.
5. Remove the nut and position the heater hose and bracket aside.



6. **NOTICE:** While removing the valve cover do not apply excessive force to the Variable Camshaft Timing (VCT) oil control solenoid or damage may occur.

**NOTICE:** If the Variable Camshaft Timing (VCT) oil control solenoid sticks to the VCT seal, carefully wiggle the valve cover until the bond breaks free or damage to the VCT seal and VCT oil control solenoid may occur.

**NOTE:** The plastic electrical connector on the VCT oil control solenoid will rotate approximately 12 degrees inside the steel housing, which is normal.

Loosen the 8 stud bolts, 3 bolts and remove the RH valve cover.

- Discard the gasket.

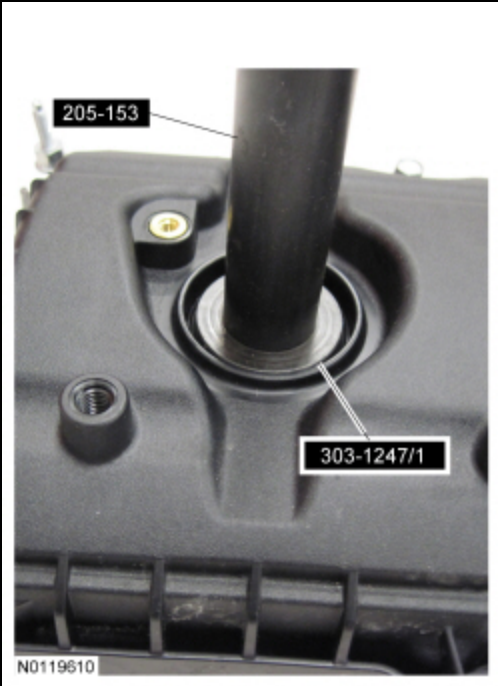
7. Inspect the 2 VCT solenoid seals. Remove any damaged seals.

- Using the Differential Bearing Cone Installer and Handle, remove the seal(s).



8. Inspect the spark plug tube seals. Remove any damaged seals.

- Using the VCT Spark Plug Tube Seal Remover and Handle, remove the seal(s).



9. Clean the valve cover, cylinder head and engine front cover sealing surfaces with metal surface prep.

**Installation**

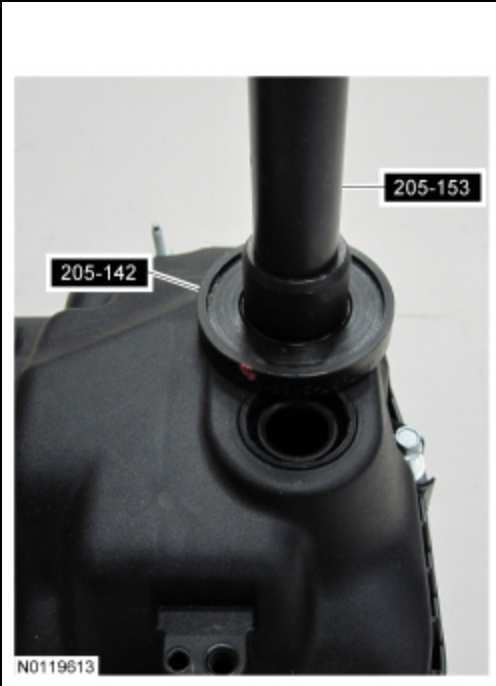
1. **NOTE:** *Installation of new seals is only required if damaged seals were removed.*

Using the VCT Spark Plug Tube Seal Installer and Handle, install new spark plug tube seals.



2. **NOTE:** *Installation of new seals is only required if damaged seals were removed.*

Using the Differential Bearing Cone Installer and Handle, install new VCT solenoid seal(s).



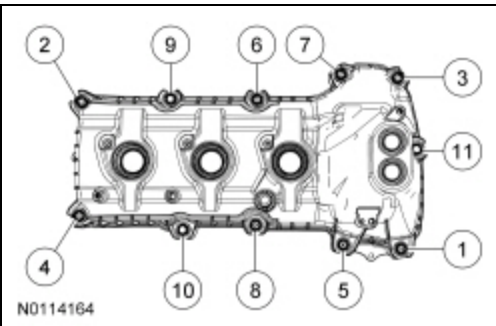
3. **NOTICE:** Failure to use the correct Motorcraft® High Performance Engine RTV Silicone may cause the engine oil to foam excessively and result in serious engine damage.

**NOTE:** If the valve cover is not installed and the fasteners tightened within 4 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Failure to follow this procedure can cause future oil leakage.

Apply an 8 mm (0.31 in) bead of Motorcraft® High Performance Engine RTV Silicone to the engine front cover-to-RH cylinder head joints.

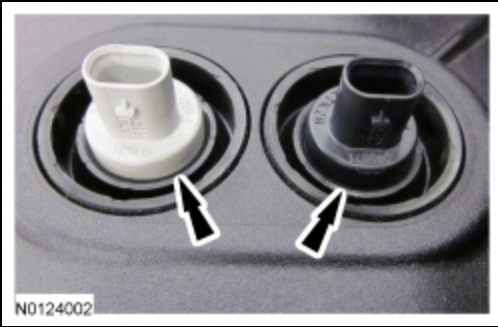


4. Using a new gasket, install the RH valve cover and tighten the 3 bolts and 8 stud bolts in the sequence shown.
- Tighten to 10 Nm (89 lb-in).



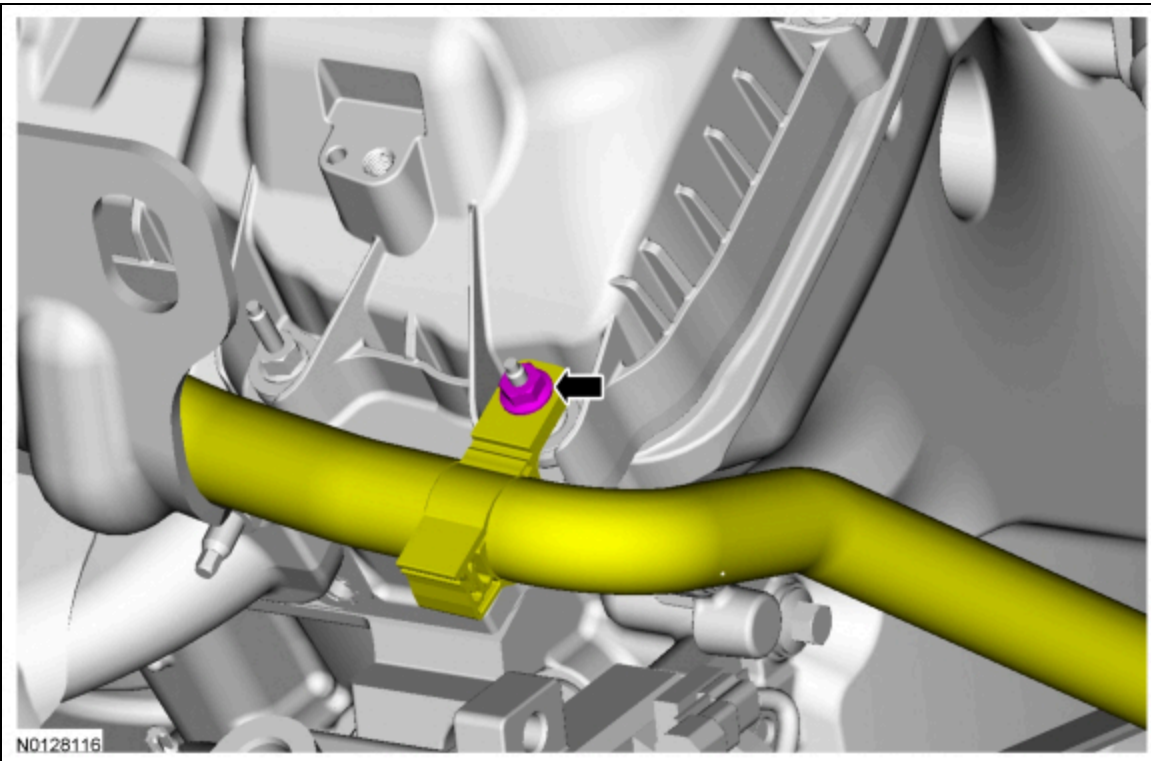
5. Ensure the VCT seals in the valve cover are below the top of the VCT oil control solenoid electrical connector or the VCT seal may leak oil.





6. Position the heater hose and bracket on the valve cover stud bolt and install the nut.

- Tighten to 8 Nm (71 lb-in).



7. Attach all of the wiring harness retainers to the RH valve cover and stud bolts.

8. Connect the 2 RH VCT solenoid electrical connectors.

9. Connect the PCV tube quick connect coupling to the RH valve cover. For additional information, refer to Section 310-00.

10. Install the RH ignition coils. For additional information, refer to Section 303-07B.